Unit 3

BASIC INFRASTRUCTURE

SURVEYING: Surveying is the art of determining relative positions of objects on or above or beneath the surface of the earth by taking measurements of distance, direction and elevation.

OBJECTIVES OF SURVEYING:

- The data obtained by surveying are used to prepare the plan or map showing the ground features.
- When the area surveyed is small and the scale to which it's result plotted is large, then it is known as plan.
- When the area surveyed is large and the scale to which it's result plotted is small, then it is called as a map.
- Setting out of any Engineering work like buildings, roads, railway tracks, bridges and dams involves surveying.

Principles of surveying: Principle 1: To work from whole to the part.

i. A number of control points are fixed in the area concerned by adopting very accurate and precise methods.

ii. The lines joining these control points will be control lines.

iii. Other measurements are made to locate points inside these control lines.

iv. This man triangles and traverse are formed first.

v. The main triangles and traverse are divided into smaller ones by using less rigorous methods.

vi. By doing so, accumulation of errors is avoided and any local errors can be easily identified.

vii. If survey work is started from a part and preceded to whole there are chances of errors getting multiplied at every stage. Hence any survey work should be from whole to part and not from part to whole.

Principle 2:

As per the principle 2, the location of a new point involves one of the following

a) Measurements of two distances.

b) Measurements of two angles.

C) Measurements of one angle and one distance

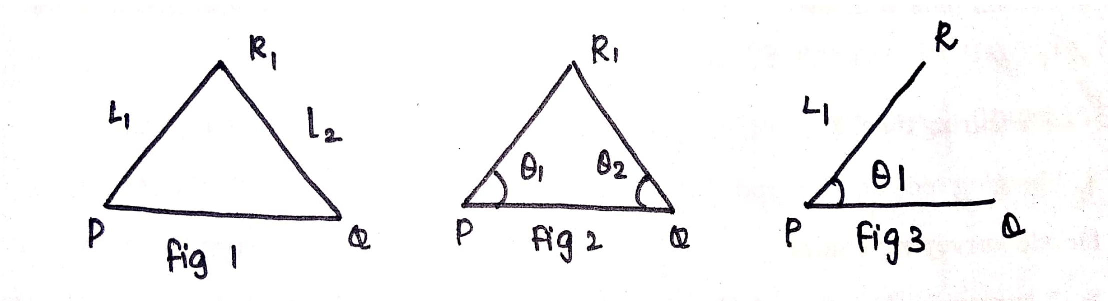


Fig.1 shows the method of locating R with reference to know length PQ by using the known distances of PR (l1) and QR (l2).

Fig.2 shows the method of locating R with reference to know length PQ by using the known angles of QPR (θ1) and PQR (θ2).

Fig .3 shows the method of locating R with reference to know length PQ by using the known distance of PR (l1) and known angle QPR (θ1)

Main divisions of surveying

i. Plane surveying

ii. Geodetic surveying

Plane surveying

- The surveying where the effect of curvature of earth is neglected and earth's surface is treated as plane is called plane surveying.
- The degree of accuracy in this type of surveying is comparatively low.
- Generally when the surveying is conducted over the areas less than 260 sq. Km, they are treated as plane surveying.
- It is conducted for the purpose of engineering projects.

Geodetic surveying

- The effect of curvature is taken into account, It is also known as "Trigonometrical Surveying".
- It is a special branch of surveying in which measurements are taken with high precision instruments.
- Calculation is also made with the help of spherical trigonometry.
- It is generally adopted by the Great Trigonometrical survey of India.

Classification of surveying

i. Land surveying
ii. Marine or Navigation or Hydrographic Surveying
iii. Astronomical Survey

Land Surveying: land survey is a one in which the relative points or objects on the earth surface is determined.

Marine surveying is one in which the relative positions of objects under water is determined.

Astronomical Survey: It is one in which observations are made to locate the heavenly bodies such as sun, moon and stars.

Classification of land surveying:

Topographical survey

It is used for determining the natural and artificial features of the country such as rivers, lakes, hills and canals.

Cadastral survey: It is used to locate additional details such as boundaries of fields, houses and other objectives.

City Survey: It is used for town planning schemes such as laying out plots, constructing streets, laying water supply and sewer lines.

Engineering survey:

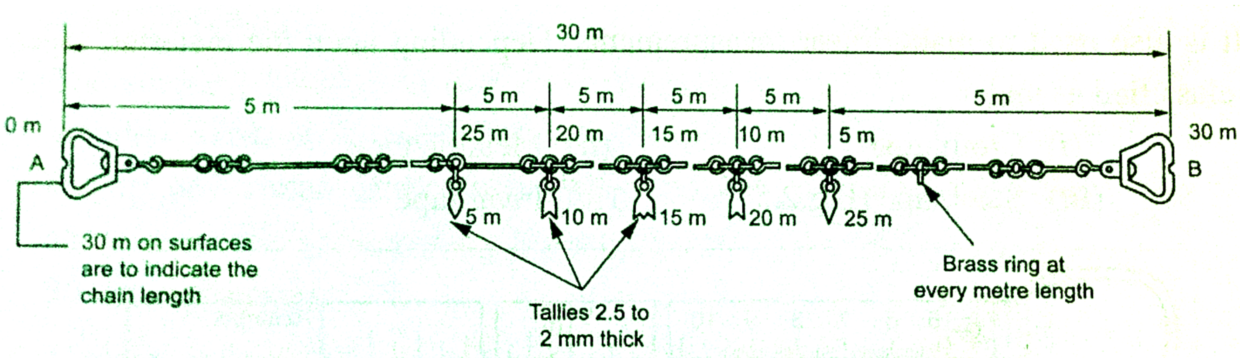
It is used to collect data for design and construction of Engineering works such as roads, railways, bridges, dams etc.,

CHAIN SURVEYING

- In chain surveying only linear distances on the field are measured.
- These distances are used to define the boundary of field and mark simple details.

PRINCIPLE:

- It is to form a network of triangles by using the distance measured.
- Better accuracy will be obtained if the triangles thus formed are nearly equilateral in shape.



Chain Surveying

Accessories used in chain surveying

- Metric chain
- Chain pins or arrows
- Pegs
- Measuring tape.
- Ranging rods or offset rod

Metric surveying chain:

- ➢ A surveying chain is a device used to measure distance between two points on the ground.
- ➢ Metric chains are available in lengths of 5m, 10m,20m and 30m.
- ➢ 20m and 30 m chain is normally used for the field of surveying.
- ➢ A surveying chain contains brass handles with brass eyebolt and collar, galvanized mild steel links and wire rings.
- ➢ In the case of 20 m and 30 m chains, brass tallies are provided at every 5 m length and indicating brass wire rings are attached at every meter length except where tallies are provided.
- ➢ The distance between the outside faces of handles of a fully stretched out chain is the length of the chain. The length of the chain like 20 m is engraved on the handles. While measuring the long distance the chain will have to be used a number of times.

Chain pins:

- ➢ Chain pins or arrows are used with the chain for marking each chain length on the ground.
- ➢ The arrow is driven into the ground at the end of each chain length is measured.
- ➢ Chain pins or arrow should be made of good quality hardened and tempered steel wire of minimum tensile strength of 70 kg/mm2.
- ➢ The overall length is 400 mm and thickness is 4 mm.

- The wire should be black enameled.
- The arrow has a circular eye at the one end and is pointed at the other end.

Pegs:

- Wooden pegs of 15 cm length and 3 cm square in section are used to establish points or the end points of a line on the ground.
- They are tapered one end and are driven into the ground by using a wooden hammer.
- About 4 cm is left projecting above the ground.

Measuring tape:

A. Cloth or linen tape

B. Metallic tape

C. Steel tape

D. Invar tape

Ranging Rod:

- It is also known as ranging pole or picket.
- Ranging rods is used for ranging or aligning long lines on the ground in field surveying.
- Ranging is a straight line means fixing a series of pegs or other marks such that they all lie on a straight line.

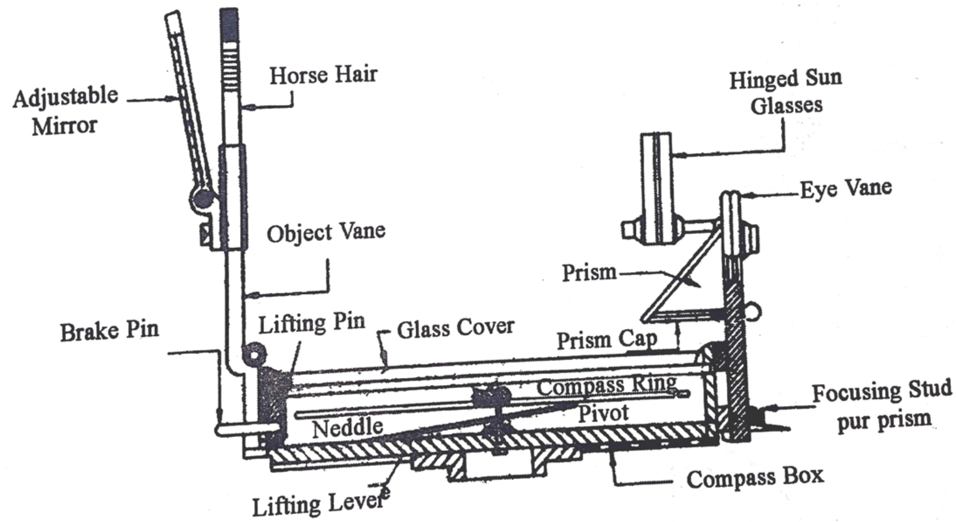
COMPASS SURVEYING

A Transverse is a multi-sided figure consisting of a series of connected lines. The lengths are measured by chain or tape and directions are identified by angle measuring instruments like compass and theodolite. This process is known as compass surveying.

Compass Survey: There are two types of compass in common use viz.,

1. The prismatic compass
2. The surveyor’s compass

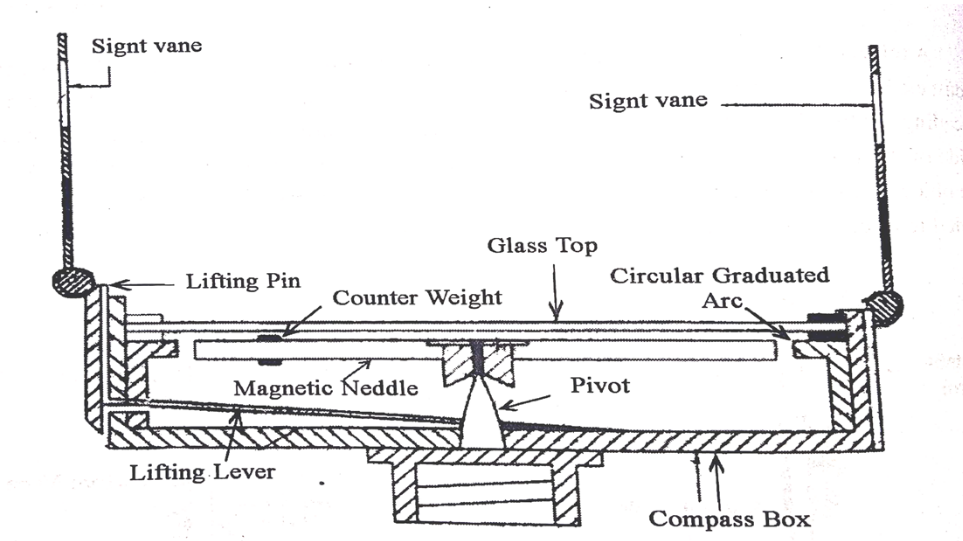
Prismatic compass:



Prismatic compass

1. It is circular in shape and its diameter varies from 85mm to 110mm.
2. A pivot is provided at the Centre of the box. It carries a magnetic needle.
3. The needle is attached to an aluminum ring which is graduated to ½".
4. A light spring break is attached to the inside of the base of damp the oscillations of the needle and bring it to rest before taking a reading.
5. A reflecting prism helps in taking reading of the angles and is protected from moisture and dust etc. by a prism cap.
6. The prism base and vertical faces are made convex which magnifies the readings.
7. The object vane is located diametrically opposite to the prism. It is hinged to the side of the box and carries a horse hair.
8. A titled reflecting mirror is provided on the side of the object vane to enable bearing of very high or low objects to be taken.
9. A metal cover is provided to enclose the compass and the object vane.

Surveyor's Compass



Surveyor's Compass

Surveyor's compass also resembles the prismatic compass except for the following points.

1. The graduated card is attached to the box and not to the needle.

2. The graduations are in quadrant bearing system having 0' at North and 90' at South and East and West interchanged.

Levelling: It is surveying method used to determine the levels.

Instruments used in levelling

- Levelling instrument like dumpy level
- Levelling staff

Method of levelling:

Method 1: It is done with only one setting of the instrument.

Method 2: When the two station points are wide apart and the instrument is set up at more than one point and the levelling is carried out.

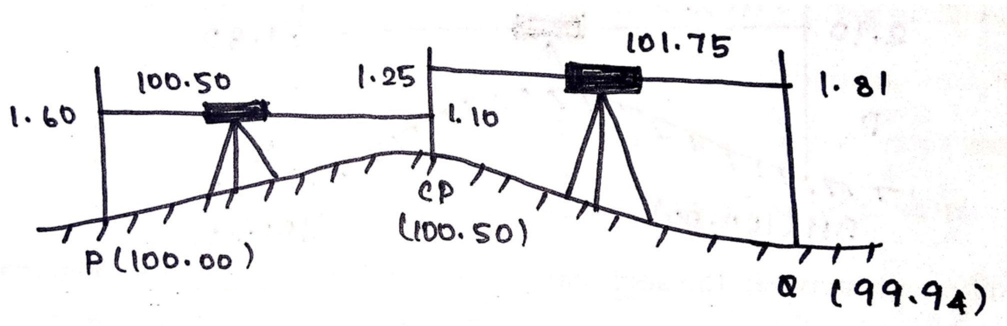
Method 1: The instrument is set up at a point between P and Q and the temporary adjustments carried out.

i. The leveling staff is held at P, the elevation of which is known already.

ii. A back sight is taken on the staff held at P. The staff is then held at Q and the foresight is taken.

Method 2:

HEIGHT OF COLLIMATION METHOD



- A changing point (CP) is established in between P and Q.
- A back sight is taken at P and a fore sight is taken at the changs point.
- The instrument is shifted to another point between the change point and Q.
- A back sight is taken at the change point and a fore sight is taken at Q.
- Any number of change points is established as required. This method is known as Height of collimation method.

First height of the instrument = Elevation of P + Back sight at P

= 100.00+1.60 =101.60m

Elevation of change point = first height of the instrument -Fore sight at change point.

= 101.60-1.10 =100.50 m

Second height of the instrument = Elevation of change point + Back sight at change point

= 100.50+1.25 =101.75m

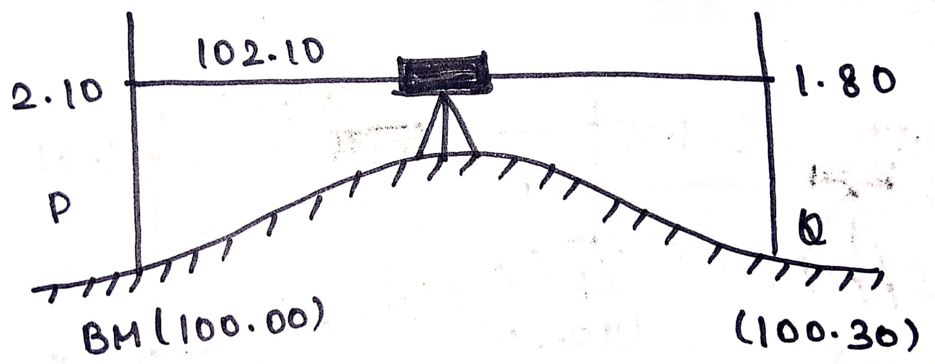
Elevation of Q= Second height of the instrument -Fore sight at Q.

= 101.75 - 1.81=99.94m

RISE AND FALL METHOD

- The staff readings of the points observed from the same setting of the instrument are compared.
- It is found whether a point is above or below the preceding point.

- If the point is above, the staff reading will be less than the preceding point. The difference between the staff readings is called rise.
- If the point is below the preceding point, the staff reading will be greater than that at the preceding point. The difference between the staff readings is termed as fall.



The difference between the staff readings at P and Q = 2.10-1.80 =0.30m (rise)

Hence, level of Q = elevation of P + Rise

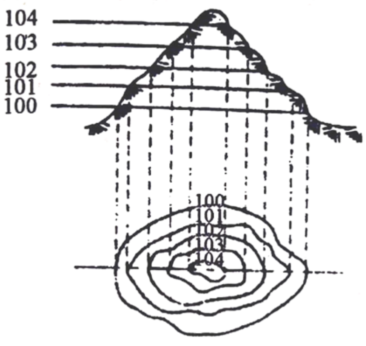
=100.00+0.30 =100.30m.

CONTOURS

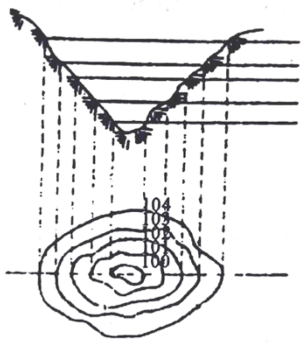
Contour is an imaginary line joining the points of same level, relative to an assumed datum. Such lines are drawn along with their levels, on the plan of an area after determining the levels of several points in that area. Such a map is known as contour map.

CHARACTERISTICS OF CONTOURS

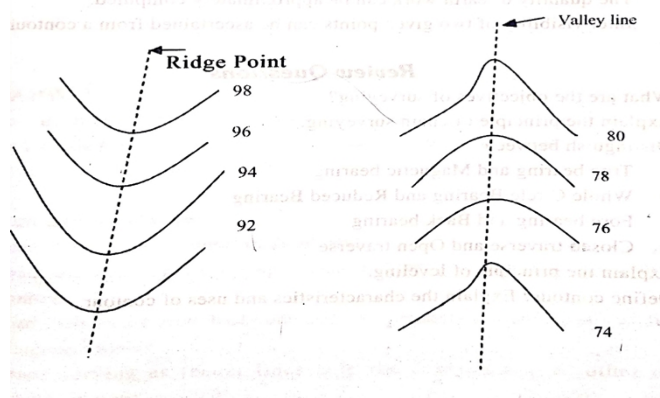
- Contour lines do not end, but close on themselves
- Widely spaced contour indicates flat surface, closely spaced contour indicates steep ground and equally spaced contour indicates uniform slope.
- Irregular contours indicates uneven surface.
- Two contours don’t interact except in the case of an overhanging cliff
- In the series of closed contours, if highest value is inside it represents a hill, if lowest value is inside it represents a valley
- Contour lines with U-shaped with convexity towards lower ground indicate ridge
- Contour lines with V-shaped with convexity towards higher ground indicate valley



CONTOUR OF A HILL



CONTOUR OF A VALLEY



USES OF CONTOURS

- Contours are used to find out the nature of the ground.
- Contours help in finding out the depth of cutting or filling for formation level of a road or railway line
- The routes of the railway , road, canal or sewer lines can be decided by using contours
- Catchment area and quantity of water flow for river can be determined from contour maps
- From the contours , it is also possible to determine the capacity of a reservoir

<u>ROAD</u>

Road is a paved path for the transport of vehicle to carry men and materials from one place to another. It is most basic and oldest form of transports.

CLASSIFICATION OF ROADS

Basically roads are classified as urban roads and non-urban roads.

Urban roads are expressways, Arterial streets, Sub arterial streets, collector streets and local streets.

Non – urban roads are National highways, State highways, Major district highways, other district roads and village roads.

COMPONENTS OR PARTS OF ROAD

Sub grade, sub base course, base course and surface course are the parts of a bituminous road.

Sub grade, base course and wearing course are the parts of a concrete road.

SUPER ELEVATION

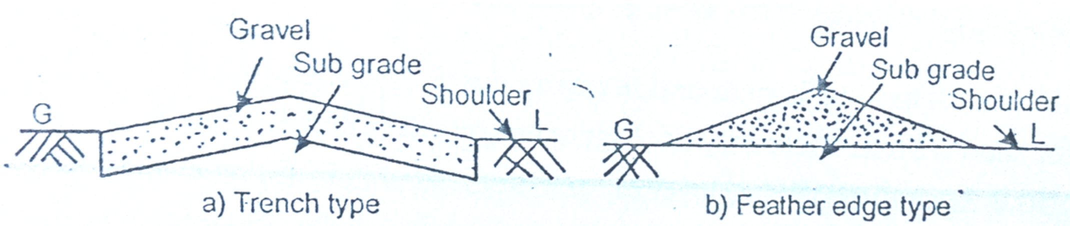
Super elevation is the additional rise in the outer edge with respect to the inner edge, provided at the curved turnings of a road. It is provided to balance the centrifugal force on the vehicle due to the curved nature of turnings and its travelling speed. If this elevation is not provided the vehicle may over shoot along the other side of the curve.

IMPORTANCE OF ROADS

- They are the most commonly used means of transport.
- They provide the cheapest form of transport after railways
- They provide faster and easy transportation
- They help in providing better law and order
- Roads provide transport during emergencies like wars, floods, famines, etc.
- They improve social, educational, commercial and economical activities and provide health facilities
- Sub grade, sub base course, base course and surfaces course are the parts of a bituminous road. Sub grade, base course and wearing course are the parts of a concrete road.

CLASSIFICATION BASED ON MATERIAL OF ROAD

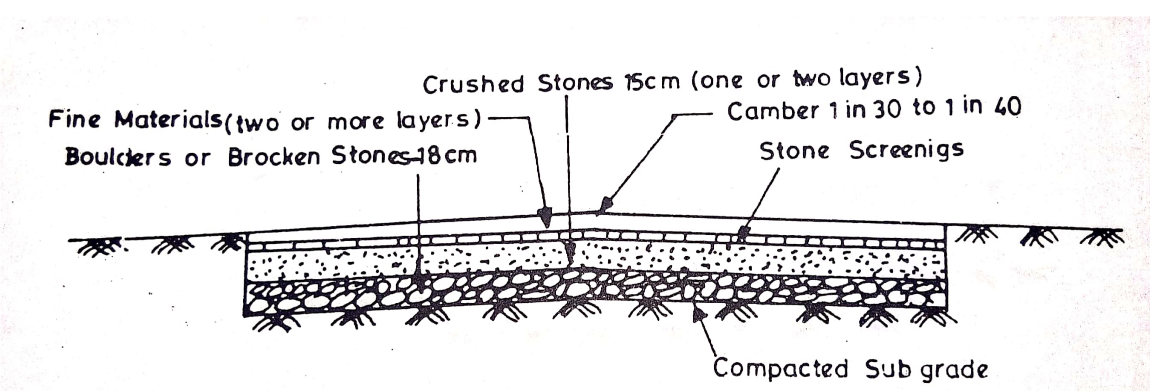
1. Earth roads: These are made from the soil available at the site. These roads are used where the traffic is very scanty. Construction cost is very cheap and their performance depends on proper maintenance and adequacy of drainage.



Earth roads

2. Gravel roads: This type of roads fall in between earth roads and metal roads. These roads are prepared by compacting a mixture of gravel and a binder material like clay or sand. This mixture is spread on the prepared sub grade and rolled to get a thickness varying from 15cm to 30cm. In the trench type, a trench is cut to the required depth and the gravel material is filled up in this trench and compacted.

3. Water bound macadam roads: These are prepared in layers. Each layer has a thickness of about 10-15cm.



Water bound macadam roads

Water bound macadam roads consists of broken or crushed aggregates mechanically interlocked by rolling and their voids filled or bound by dust and water. It is generally used as sub base or base course.

Construction used for Water Bound Macadam Road.

Preparation of sub grade:

This is provided to the desired elevation and cross section. Then rolling is done for achieving compaction.

Preparation of sub base:

On the sub base a layer of broken stones is laid to get a thickness of about 15cm. Gravel and screening materials are then spread over this. Then rolling is done adding water.

Sub base cum wearing course: This course is about 10cm thick. About 4cm size broken stones are spread over the prepared sub base. This is then rolled dry for proper interlocking of the stones. Enough filler materials are added to fill the surface voids.

Rolling is then done to get a thoroughly compacted surface. Then water is sprinkled over this surface and rolled again

Blind age material application

Blind age material like soft murum, kankar or fine quarry dust is spread uniformly above the base course for about 12mm thickness. Sufficient water is then added on this layer to form slurry. This is then rolled. Then a layer of sand is spread uniformly on the above surface to a thickness of 6-10mm.

The surface is then cured for 1-2days before allowing the traffic; the surface is kept wet for at least 7 days.

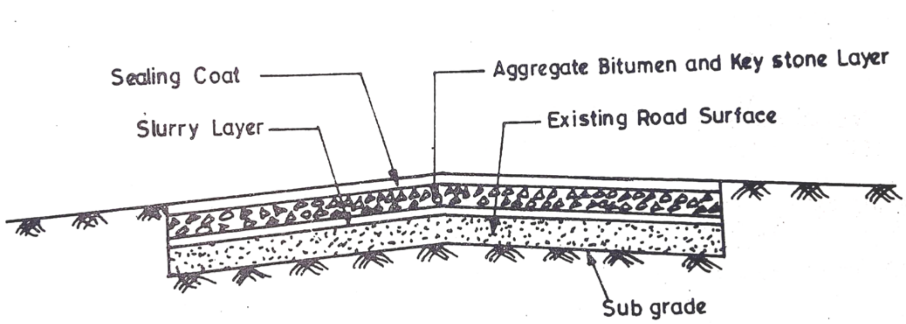
Advantages of Water Bound Macadam road

- It is stronger than earthen road and gravel road
- It can be used as base course and sub base course
- It is denser due to interlocking of the aggregates and hence can take more loads

Disadvantages of Water Bound Macadam road

- It doesn't has any strong binder material and hence cannot be used as wearing course

4. Bituminous road

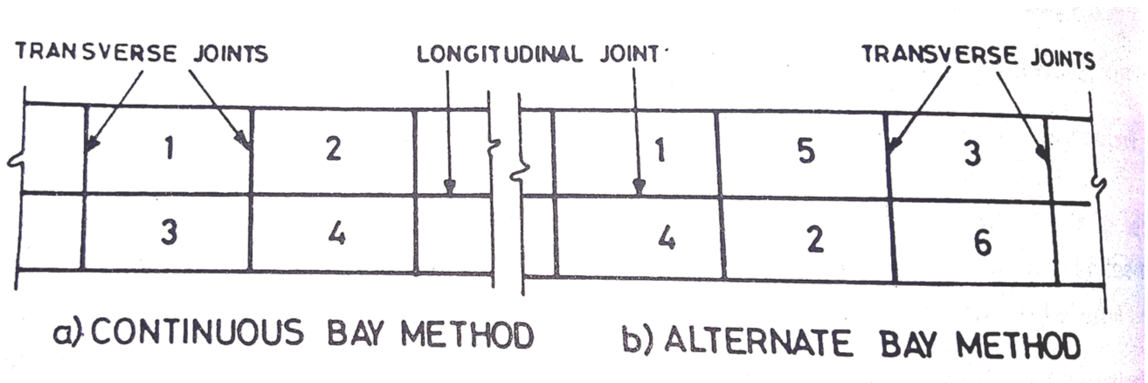


Here bitumen is used as the binding material. This type is also called as a black top road. The bituminous material used can be classified into two-groups-Asphalt and tar. Bitumen containing some inert materials or minerals is called Asphalt. Asphalt occurs in nature and is called rock asphalt. Tar is a residual product of the destructive distillation of coal. Road tar is coal tar treated to conform to specifications to make it suitable for road applications.

5. Concrete road

The following steps are followed in the construction of concrete road

- First the sub grade is prepared and cleaned of dirt and loose materials.
- Then the sub base course of concrete is laid in layers up to required thickness and compacted and levelled
- After 24 hours to give sub grade time to set, it is prepared for laying base concrete
- The form works are placed and base concrete is laid in layers and compacted and levelled
- Then the wearing course is laid with rich mortar and allowed to cure for 28 days
- The entire length and width of the road is divided in to bays and the order of construction of bays is as follows:
- In alternate method the alternate bays (as shaded in diagram) are laid first and then remaining bays are laid. In continuous method the width of road is divided in to half and each width is laid one after another.



Concrete road

Advantages of concrete roads

- They provide excellent riding surfaces and pleasing appearances
- They provide better visibility
- They can withstand heavy traffic
- They are not much damaged during rain and neither by oils
- The contents of concrete roads can be mixed under normal temperatures
- They are more durable and hence maintenance of concrete roads is easier

Disadvantages of concrete roads

- Cost of concrete roads is greater than the bituminous roads
- They cannot be opened for traffic after construction as they require curing period
- Stage development is not possible in concrete roads
- Joints in concrete roads are weaker than other parts and may be easily damaged

COMPARISON BETWEEN BITUMINOUS CONCRETE ROAD AND CEMENT CONCRETE ROAD

No.	Item	Bituminous concrete road	cement concrete road
1.	Compaction	It is achieved by rolling with usual precautions.	It is done by the use of vibrators
2.	Curing	Not necessary	Necessary
3.	Formwork	Not necessary	Required for edges and near joints
4.	Initial cost	Less	More
5.	Joints	Not to be provided	To be invariably provided.
6.	Maintenance cost	More	Less
7.	Method of placing	The premix is prepared in hot mix plant and it is to be placed in hot condition	The concrete is prepared by mixing cement, aggregates and water and it is to be deposited before cement starts to difficult set. No heating is required.
8.	Opening of underground trenches	Can be easily opened up	Difficult to open
9.	Opening to traffic	After 24 hours	After 15 to 20 days
10.	Salvage value	Practically nil	Can serve as a base or foundation for the new bituminous surface
11.	Tractive resistance	More	Less
12.	Useful life	Less	More
13.	Utility	Less durable for heavy traffic, cost of repair and maintenance more	More durable and reliable for heavy traffic
14.	Visibility at night	Bad	Good
15.	Wear of surface	More	Less

BRIDGES

A Bridge is a structure constructed to provide passage for a road or railway over an obstacle such as river, valley etc. without disturbing or closing the way beneath. The passages include pedestrians, a canal or a pipe line and the obstacles include a road or a railway

COMPONENTS OF A BRIDGE:

The various components of a bridge consist of: 1.Foundation, 2. Piers, 3. Abutments, 4. Bank connections, 5. Approaches, 6. bearings, 7. Decks, 8.hand.

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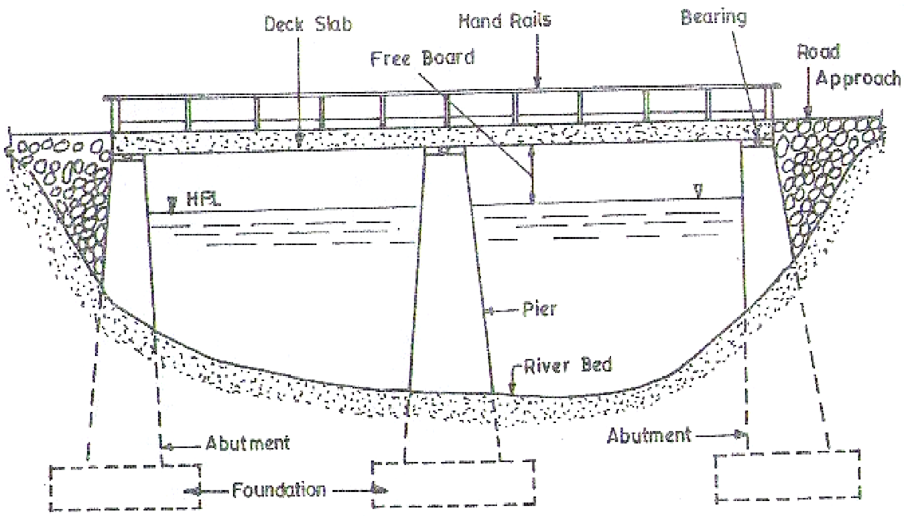


Components of Bridge

FOUNDATIONS:

The load of the bridge and the traffic are transmitted to the subsoil through piers, abutments, etc., usually deep foundation is preferred for piers and abutments. However the subsoil is sound shallow foundation are provided.

PIERS:

Piers are the intermediate supports to the superstructure. They divide the length of the bridge to different spans keeping economy into consideration and transfer the load from deck to the foundation.

ABUTMENTS:

Abutments are the end supports to the super structure. They function as a pier and as well support the backfill and also allow construction of bank connection.

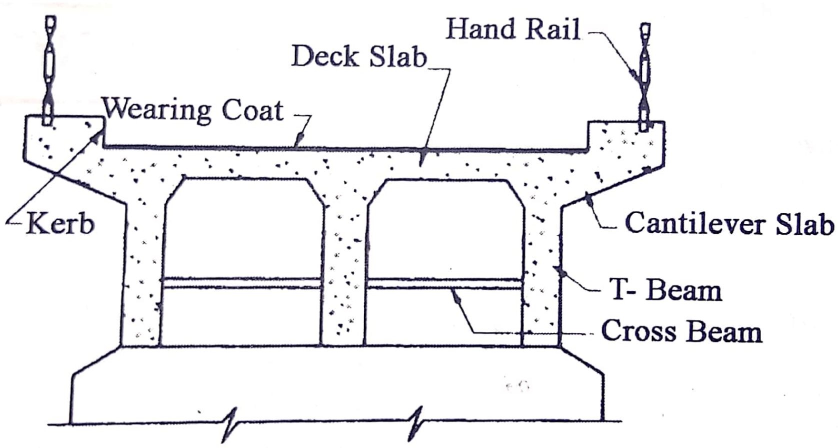
BANK CONNECTION:
Bank connections such as wing walls and return walls provide a firm connection between the bridge abutments and the road approach. They function as retaining walls and also direct the river water into the spans of the bridge.

APPROCHES:
That portion of the road or railway constructed to reach the bridge from the proposed alignments is known as the approach of the bridge.
BEARINGS:
Bearings are parts of a bridge provide to transmit the load from the superstructure to the substructure in such a manner that the bearing stresses induced in the substructure are within permissible limits and also to allow for linear and angular movements.

DECK:
Deck is the top portion of the bridge provided to form roadway or railway. It spans continuously from abutment to abutment over the piers. It may consist of a slab, girders, trusses, arches, etc..,

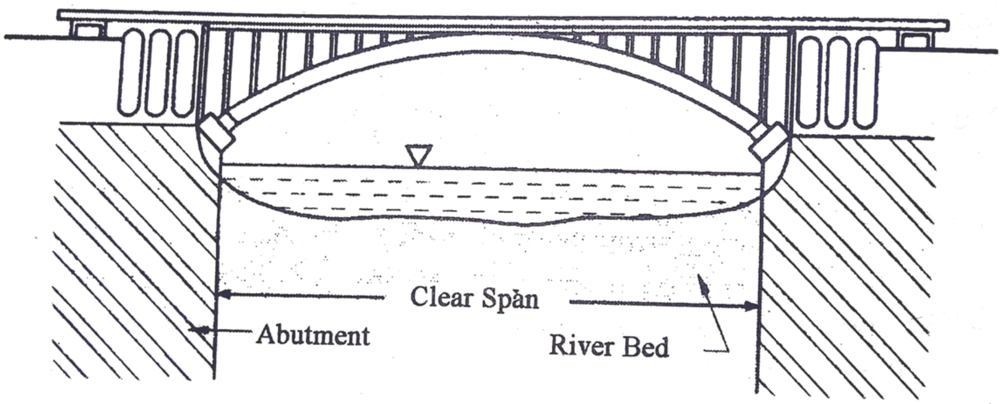
HAND RAILS:
Hand rails, parapet or guard stones are provided on either sides of the deck along the road way to safeguard the moving vehicles and the passages on the bridge deck.

SOME POINTS IN LOCATING A BRIDGE ACROSS A RIVER
- The span of the bridge should be perpendicular to the direction of flow of the river as much as possible.
- The width of the river at the bridge crossing should be of least width.
- The longitudinal direction of the bridge should be as much as parallel to the direction of the passage

TYPES OF BRIDGE:
T-beam Type Bridge


This is a type of RCC Bridge and it is used for span up to 20m. The T-beam is simply supported over abutments or piers. It is cheaper and requires less quantity of materials. Over concrete weathering layer is laid on which traffic moves. When this layer wears or worn out it is replaced by another layer. Generally in RCC abutments and piers are constructed with brick or stone masonry.

Steel Arched Bridge

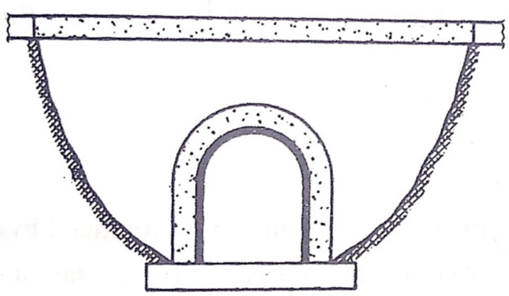


The bridge having their super-structure consisting of steel or iron member in any structural form are known as iron or steel bridge. Steel arch bridge is one of the steel bridges.
It is more suitable for deep gorges. The steep rocky banks provide a natural abutment and could take heavy thrust. It gives more pleasant appearance. It may be designed with hinges or not.

Culvert

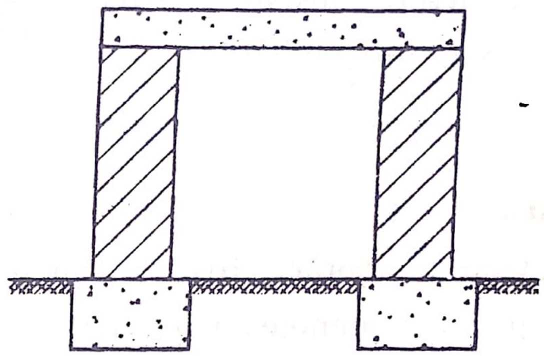
Culverts are cross drainage works with clear span less than 6m. The various types of culverts are as follows.

Arch culvert



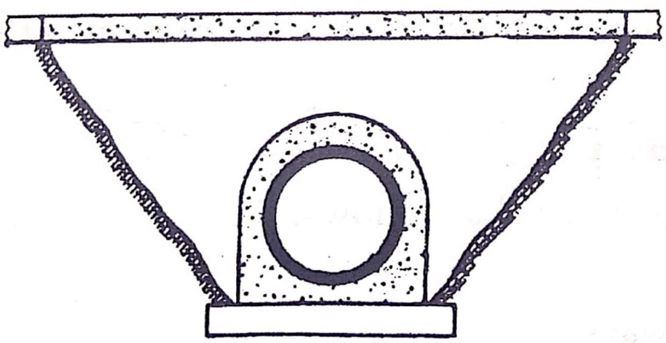
It consists of abutments, wing walls, arch, parapet and the foundations. Masonry arches are commonly used. The size of arched opening can be adjusted to carry the required quantity of water. The abutments or piers carry the thrust of arch and may be made of masonry or concrete.

Open slab culvert



It is used where bed of canal is sufficiently firm and the site of culvert is on straight level reach. A slab culvert is normally used up to span of 3m.

Pipe culvert



It is suitable when the discharge in the stream is very small or when sufficient headway is not available. Generally the pipe of diameter not less than 300mm is used. The faces of pipes may be surrounded by pitching to protect the embankment from the action of spring water.

Box culvert

It is a culvert consisting of rectangle or square opening having their floor top slab constructed monolithically with abutments and pipes.

DAM

Dam is a structure used to store water. It is constructed across a river preferably at a valley. It is constructed with water tight materials like stone masonry, RCC, or steel.

The parts of a dam are:

Reservoir, embankment, impervious core or lining, foundation, spill ways, sluice gates to control flow of water.

PURPOSES OF DAMS

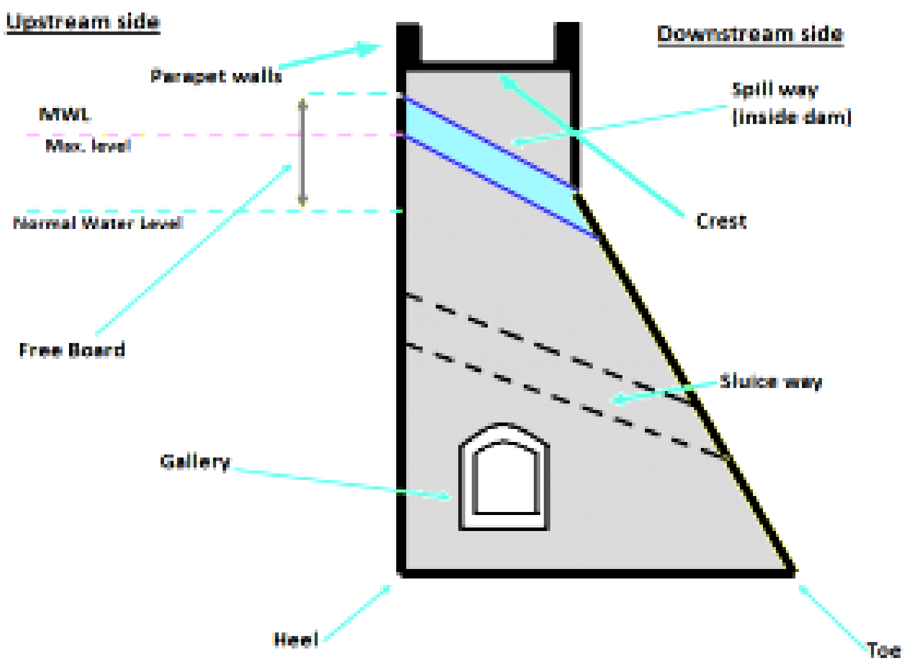
- To store and control the water for irrigation
- To store and divert the water for domestic uses
- To supply water for industrial uses
- To develop hydro electric power plants to produce electricity

- To increase water depths for navigation
- To create storage space for flood control
- To preserve and cultivate the useful aquatic life
- For recreational purposes

FACTORS TO BE CONSIDERED IN SELECTION OF SITE FOR DAM

- The dam site should not be an earthquake prone zone or seismically active zone.
- Suitable foundations should be available at the site selected for a particular type of dam. For gravity dams, sound rock is essential. For earth dams, any type of foundations is suitable with proper treatment.
- The river cross-section at the dam site should preferably have a narrow gorge to reduce the length of the dam.
- The general bed level at dam site should preferably be higher than that of the river basin. This will reduce the height of the dam and will facilitate the drainage.
- A suitable site for the spillway should be available in the near vicinity.
- Easily availability of construction materials and man power an ease in access.

Different parts & terminologies of Dams:



Dam-Parts in a typical cross section

Crest: The top of the dam structure. These may in some cases be used for providing a roadway or walkway over the dam.

Parapet walls: Low Protective walls on either side of the roadway or walkway on the crest.

Heel: Portion of structure in contact with ground or river-bed at upstream side.
Toe: Portion of structure in contact with ground or river-bed at downstream side.
Spillway: It is the arrangement made (kind of passage) near the top of structure for the passage of surplus/ excessive water from the reservoir.
Abutments: The valley slopes on either side of the dam wall to which the left & right end of dam are fixed.
Gallery: Level or gently sloping tunnel like passage (small room like space) at transverse.. These may also be used to accommodate the instrumentation for studying the performance of dam.
Sluice way: Opening in the structure near the base, provided to clear the silt accumulation in the reservoir.
Free board: The space between the highest level of water in the reservoir and the top of the structure.
Dead Storage level: Level of permanent storage below which the water will not be withdrawn.
Diversion Tunnel: Tunnel constructed to divert or change the direction of water to bypass the dam construction site. The hydraulic structures are built while the river flows through the diversion tunnel.

CLASSIFICATION OF DAMS:

1. Based on its use, the types of dams are:

Storage dam, diversion dam, detention dam, hydro electric power producing dam, irrigation dam, etc

2. Based on material used for construction, the types of dams are:

Earthen dam, rock-fill dam, timber dams, steel dams and concrete dams

3. Based on construction and action of hydraulic forces, the types of dams are:

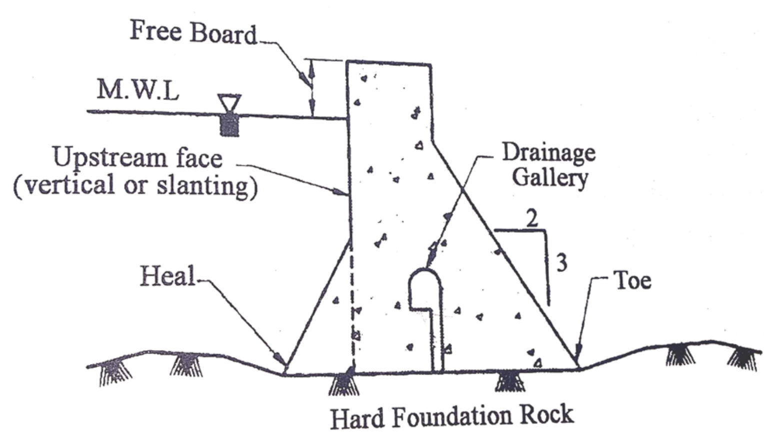
Gravity dams, arch dams and buttress dams

4. Based on its hydraulic design, the types of dams are:

Over flow dams and non-overflow dams

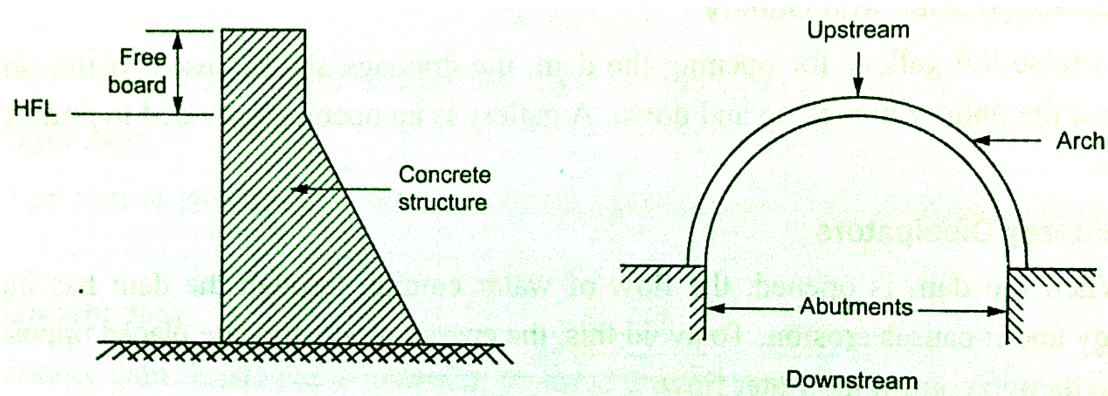
Based on construction, dams are classified as the following:

Gravity Dam:



They consist of a solid embankment made of masonry or concrete which is used to hold the water. The weight of the embankment should balance the lateral pressure of stored water. Hence a dense material like concrete is used in their core. They require large quantities of material and may prove to be costly for storing large quantities of water.
They require less structural components like foundation. They require large quantities of materials and may prove to be costly for storing large quantities of water.

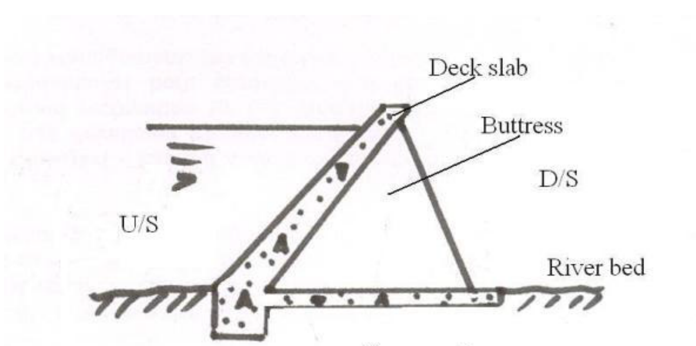
2. Arch dams:



Arch dams

They consist of an arch like curved structure which holds the water with the concave side towards of downstream side. This arch is supported by abutments at the end and foundations at their base. The curvature of the arch and its reaction at abutment balances the water pressure. This arch can be constructed of concrete or steel.
It requires less material as its cross section is thinner than that of gravity dam. Hence it is cheaper. It is preferable for great heights. Arch dams are preferable at valleys with narrow width. Arch dams cannot be used as very large dams.

3. Buttress dams: They are similar to gravity dams but with thinner sections and intermediate support members called buttress. Buttresses are provided in the downstream direction to balance the pressure of water. Buttresses dams can be constructed on soils with weak bearing capacity.



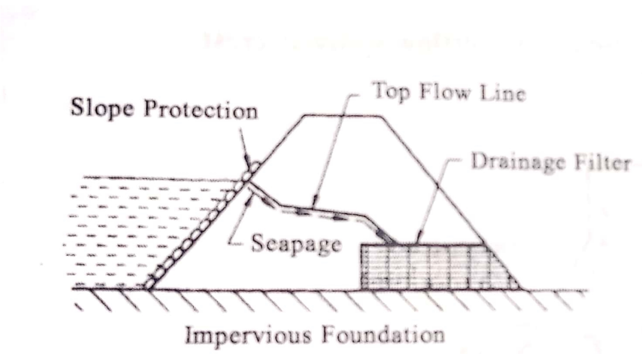
Buttress Dam

BASED ON THE MATERIAL USED FOR CONSTRUCTION, DAMS ARE CLASSIFIED IN THE FOLLOWING:

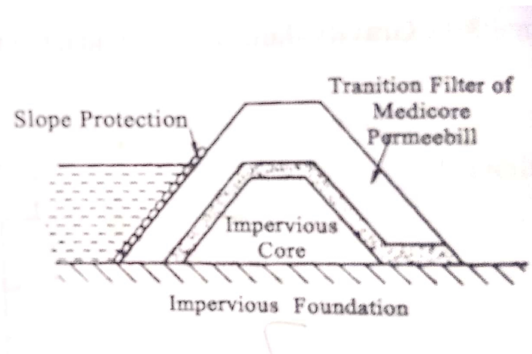
1. Earth dams: These are well suited and economical when the height of the dam is medium and can be constructed on any type of foundation. The soil is laid in this layer, added with the proper moisture content and compacted with rollers to develop the require strength.

The upstream face of the dam holding the water is provided with stone revetment. It is not made water tight.

Filters are provided so that the seepage water does not carry any material of the dam. Earth dam last long if properly designed constructed and maintained.



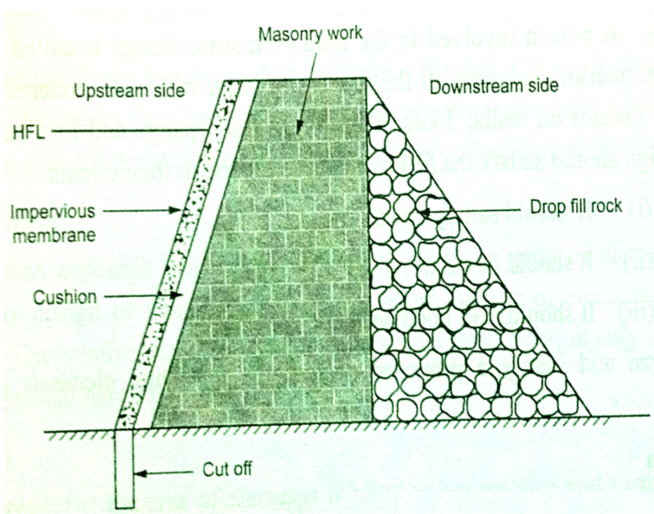
Homogeneous earth dams



Zoned earth dam

2. Rock fills dams: In places, where good rock is available in plenty, this type can be adopted. This type is suitable for moderate heights. A concrete slab is provided on the upstream side.

The dam section consists of dry rubble stone masonry on the upstream side and loose rock fill on the downstream side. If rock foundation is available, then the settlement of the dam is minimum. This type has better resistance to earthquake forces because of its flexible nature.



Rock fills dams

ADVANTAGES OR USES OF A DAM

- A dam is used to store water for usage in summer
- The water stored is used for drinking, irrigation, grow fishes, recreation and generate hydro electric power
- Dams are also used to avoid floods, to divert the flow of water and also to increase river depth for navigation.

DISADVANTAGES OF CONSTRUCTION OF DAMS

- Dams cause a large area of land and forest to submerge along its sides and affect people and wildlife
- Dams have possible risk of failure resulting in huge loss of life and property due to any unexpected reasons like heavy rains, floods flood, land slide, earthquakes, etc.
- Dams are too much expensive and require a lot of resource and time to construct.

WATER SUPPLY

AS PER IS 1172: 1971, WRITE THE DETAILS OF DOMESTIC CONSUMPTION OF WATER.

- The average per capita demand for domestic consumption of water is 135litres/capita/day
- The demand for gardens is 3.5litres/sq.m/day

SOURCES OF WATER

- Precipitation in form of rain, snow, dew, hail
- Surface water sources like lake, pond, river, and stream
- Ground water sources such as springs, infiltration galleries, well, ground well
- Water from any source contains impurities in it and need to be treated.
- Water treatment plants treat the water in large volumes.
- Filtering, sedimentation, aeration, reverse osmosis, disinfection like chlorination, UV rays treatment, etc. are the various method employed in water treatment plants to treat water from impurities. Then the water is sent to consumers through pipelines.

WATER SUPPLY CAN BE DONE IN THE FOLLOWING METHODS:

1. Continuous water supply: Where water is made available through the pipelines all along the days for 24 hours.

2. Intermittent water supply: Where water is made available through the pipelines only particular interval of times, like from morning 10.00 A.M. to evening 4.00 P.M.

Water has to be sent from treatment plants to the consumers through the pipelines using some force.

That force may is gravity or pressure created by pumps or combination of both. Therefore the types of water distribution system are gravity system, pumping system and dual system.

SYSTEMS OF WATER SUPPLY

Water can be supplied within the city or town by the following methods.

1. Gravity system

This system can be advantageously adopted if there is naturally elevated ground in that area. The distribution reservoir is constructed in the elevated ground in that area. The distribution reservoir is constructed in the elevated ground and the water from the reservoir can be led through pipes to supply water by gravity to the city and town.

2. Pumping system: The water from the water source (stream, rivers, ground water, etc.) is directly pumped into distribution main. A pump house is constructed and the number of pumps required depends upon the quantity of water to be supplied.

3. Combination of pumping and gravity system

From the water source, the water source is first pumped into a storage reservoir, or overhead tank. Then the water from the overhead tank is supplied to the mains by means of gravity flow.

The water pipeline networks may be of the following types: Dead end system, Grid iron system, Circular system and Radial system.

QUALITY REQUIREMENTS OF WATER:

1. Physical quality requirements:

They are important for satisfaction of the user and probably does not result in much harm

- Colour of the water should be from 15-25 when measured in cobalt scale
- Turbidity of the water should be less than 10NTU.
- The temperature of the water is preferred from $10^0C – 15^0C$
- The water should not contain any taste or odour

2. Chemical quality requirements:

The chemical impurities result in various disorders of the parts of the human body. The impurity is measured in ppm- parts per million or mg of impurity per litre of water. The maximum permissible limits of some impurities are given below.

Total solids (suspended & dissolved)	< 50-100ppm
Chlorides	< 250ppm
Sulphates	< 250ppm
Magnesium	< 125ppm
Iron	< 0.3ppm

pH value should be within the limit of 6.5-8.5

Hardness should be within 75 to 115 units of equivalent $CaCO_3$

3. Bacteriological quality requirements:

Pathogens or harmful micro organisms in water will result in various diseases like cholera, etc. These impurities should < 1 colony of M.P.N coli form bacteria/100ml

METHODS OF TREATING WATER

1. Filtration: Filtration is the process of screening water by passing it through a filter medium with pores, like sand. It is the most common method to clean water from solid particles.

2. Sedimentation: Sedimentation is a process of cleaning the water by allowing the impurities to settle as sediments. Some chemical agents are used to increase the sediment size and weight by bonding them together. Then this is called flocculation.

3. Aeration: Aeration means allowing the air to pass into the water. The oxygen air in the air mixes with water and decomposes the chemical impurities in water and cleans it.

4. Reverse-osmosis: Reverse osmosis is a process similar to filtration using a semi permeable membrane. A semi permeable membrane allows the impurities in water to pass through it along in only one direction. It requires some pressure and hence energy is required to drive this process. Hence it is costlier but most effective.

5. Chlorination: Chlorination is defined as the process of adding chlorine to water. It results in disinfection of water and makes it fit domestic consumption. It is the most common method and also economical method. It results in cleaning the water from bacteria and other micro organisms.

6. Ultra-violet rays: Ultra-violet rays are used to kill the pathogens or germs in the water which cause diseases. Even though they are very effective they are very costly.

METHODS OF CHLORINATION:

The different methods of chlorination are: pre-chlorination, post-chlorination, double-chlorination, break point chlorination and super chlorination.

1. Pre-chlorination is done before treating the water. The amount of chlorine dose is such that the residual chlorine or excess chlorine in the water is from 0.1 to 0.5ppm.

2. Post-chlorination is done after treating the water. The amount of chlorine dose is such that the residual chlorine or excess chlorine in the water is from 0.1 to 0.2ppm.

3. Double -chlorination is done during the treatment of water in two different stages. The amount of chlorine dose is such that the residual chlorine or excess chlorine in the water is from 0.1 to 0.2ppm.

4. Break Point-chlorination means optimum chlorine requirement. The amount of chlorine dose is such that the residual chlorine or excess chlorine in the water is from 0.2 to 0.3ppm.

5. Super -chlorination means chlorine dose is more than normal dose. The amount of chlorine dose is such that the residual chlorine or excess chlorine in the water is from 0.5 to 2ppm. This is ring rainy done during the times of epidemic or spreading of diseases during rainy season.

RAIN WATER HARVESTING

It may be defined as process of storing the natural infiltration of rainwater or surface run off into the ground by some artificial methods.

The methods suggested are recharge through pits, trenches, bore-wells shafts by directly diverting runoff water into existing or disused wells or conserving the rain water by artificial storing and using the same for human use. The choice and effectiveness of any particular method is governed by hydrological and soil conditions and ultimate use of water.

Benefits of Rain Water Harvesting (RHW)

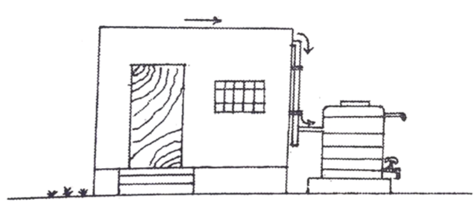
- RWH helps to conserve and augment the storage ground water aquifers thereby improving the ground water table.
- Over extraction of ground water in coastal areas leads to saline water intrusion. The recharging of ground water aquifer helps arrest this saline water intrusion.
- The quality of ground water in many pockets is either poor or not potable due to the presence of unfavorable salt contents. Continuous recharge of ground water using rainwater helps to dilute these contents and improves the ground water quality considerably.
- The buildings, which are constructed on clay soil formations, are prone to develop cracks in walls and structures during dry periods. As RWH helps to sustain the moisture level in the sub soil, such adverse conditions could be avoided.
- The scope of natural recharge has become minimal in urban areas. RWH helps recharging the ground water.
- The RWH assures more continuous, more reliable access to water.
- RWH system collects water within the accessible distance of the places of its use.

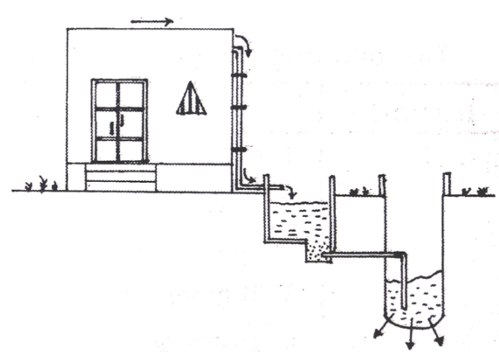
Methods of rainwater harvesting in urban cities

Broadly the rain water can be harvested by two methods

(a) Store the rain water in containers above or above ground or below ground.

(b) Recharge into soil for withdrawal later by ground water recharging basis.



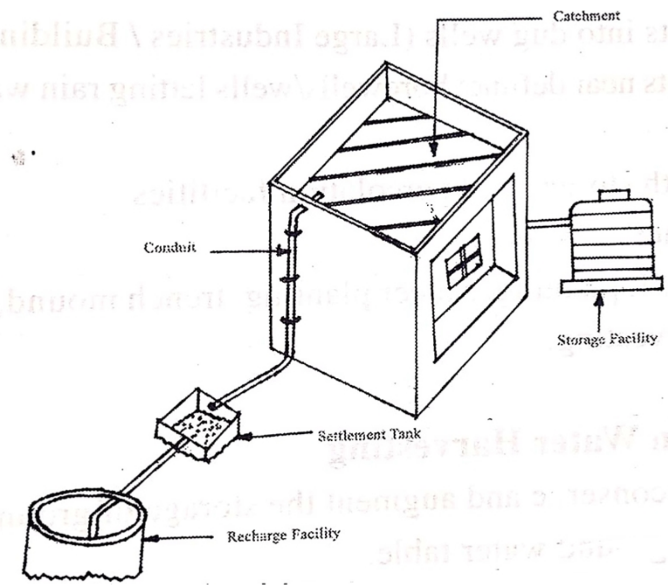


(a) Rainwater stored in tanks **(b) Rainwater recharged into the ground**

ELEMEMTS OF TYPICAL RAIN WATER HARVESTING

Any rain water harvesting will have three elements.

1. Catchment area
2. Conduits or connecting system
3. Storage and recharge system



The following places will be utilized as catchment area for collecting rainwater.

- ✓ Roof tops of the building
- ✓ Compound area of the buildings
- ✓ Areas adjoining to tanks, ooranies and temple tanks.
- ✓ Rocky surfaces or hill slopes.

The following are being utilized for transporting water from catchment areas to storage points.

- Open channels
- Closed channels
- PVC pipes,Cement pipes
- Metal pipes etc.
- Storage and Recharge structures

The rain water will be collected, stored and recharged by the following methods.

- ✓ Storage containers connected to the buildings
- ✓ Recharge trenches
- ✓ Recharge pits
- ✓ Recharge inlets into dug wells
- ✓ Recharge points near wells and bore-wells letting rain water in open wells or bore wells
- ✓ Check dams across streams
- ✓ Conventional pit planting, saucer planting, trench mound, staggered trenches and open are harvesting.

Two Marks

1. Define surveying (May 2013,Jan 2015,May 2014, May 2019)

Surveying is the art of determining relative positions of objects on or above or beneath the surface of the earth by taking measurements of distance, direction and elevation.

2. State the principle of surveying? (nov 2017)

The two general principles of surveying are:
(i) To work from the whole to the part
(ii) To fix the positions of new stations by at least two independent processes.

3. What is the function of surveying?(Nov 2018)

- ✓ Topographical maps showing hills, rivers, towns, villages, forests etc. are prepared by surveying
- ✓ Engineering map showing the position of engineering works like road, railways, buildings, dams, canals, etc. are prepared through surveying.

4. Classify surveying based on application of surveying. (may 2014) or Give the classification of surveys (nov 2017, May 2019)

- ✓ Engineering survey, Military survey, mine survey, Geological survey and Archaeological survey

5. Differentiate bridge and dam.(Nov 2018)

Bridge: It is structure that allows people or vehicles to cross any obstacles such as river or canal or railway etc.
Dam: It is barrier constructed across a river valley for water storage and to hold back the water flow.

6. How highways are classified according to their important.(Jan 2010)

- Basically roads are classified as urban roads and non-urban roads.
- Urban roads are expressways, Arterial streets, Sub arterial streets, collector streets and local streets.
- Non – urban roads are National highways, State highways, Major district highways, other district roads and village roads.

7. What are the components of parts of a bridge.(April 2010 ,Nov 2017, May 2019)

The components of the bridge are substructure and super structure. Substructure consists of foundations, abutments and piers, wing walls and bearings. The super structure consists of approaches, deck and parapet walling railing. Bridges are made of various materials such as timber, steel, RCC, masonry, pre stressed concrete bridges and composite materials.

8. What is local attraction? How it affects the survey instruments.(April/May 2014)

As the compass needle is a magnetized one, it is liable to get seriously deflected from its normal position if external magnetic influence from materials such as magnetic rocks or iron ore deposits, steel structures, railway lines, iron lamp posts, transmission towers, etc. is present. Such disturbances due to the presence of magnetic field are called the local attraction.

9. How are bridges classified according to the span?(April 2010)

- Culverts - span less than 6m
- Minor bridges – span from 6m to 30m
- Major bridges – span from 30m to 120m
- Long span bridges- span above 120m

10. Why gravity dam is so called? (Nov 2011)

A gravity dam is a massive sized dam fabricated from concrete and designed to hold back large volumes of water. By using concrete, the weight of dam is actually able to resist the horizontal thrust of water pushing against it. This is why it is called a gravity dam

11. What are the different sources of water supply(Jan 2013)

- Precipitation in form of rain, snow, dew, hail
- Surface water sources like lake, pond, river, and stream
- Ground water sources such as springs, infiltration galleries, well, ground well.

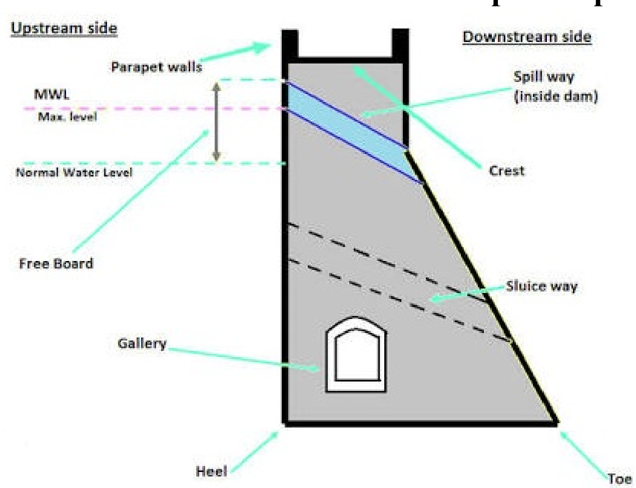
12.As per IS 1172-1971 write the details of domestic consumption of water.(Jan 2011)

- The average per capita demand for domestic consumption of water is 135litres/capita/day
- The details of domestic water consumption is India (IS: 2065-1972) are as follows:

Drinking	- 5 litres/capita/day
Cooking	- 5 litres/capita/day
Bathing	- 55 litres/capita/day
Clothes washing	- 20 litres/capita/day
Utensil washing	- 10 litres/capita/day
House washing	- 10 litres/capita/day
Flushing of latrines	- 30 litres/capita/day
Total	- 135 litres/capita/day

- The demand for gardens is 3.5litres/sq.m/day.

13.Draw a simple line sketch of a dam and mention its component parts.(Nov 2016)



14. Give examples for various instrument used in surveying.((April/May 2016,Nov 2016)

a. Chain surveying
b. Compass surveying
c. Plane table surveying
d. Theodolite surveying
e. Tachometric surveying
f. Ariel surveying
g. photographic surveying

15. What is difference between pier and abutment?(Jan 2016) or Define piers.(April 2012)

- Piers are the intermediate supports to the superstructure. They divide the length of the bridge
- Spans keeping economy into consideration and transfer the load from deck to the foundation.
- Abutments are the end supports to the super structure. They function as a pier and as well support the backfill and also construction of bank connections.

16. Write any two purpose of a dam?(April/May 2016) or Mention the purpose for which a dam is constructed.(Nov 2016)

- To store and control the water for irrigation
- To store and divert the water for domestic uses
- To supply water for industrial uses
- To develop hydro electric power plants to produce electricity
- To increase water depths for navigation
- To create storage space for flood control
- To preserve and cultivate the useful aquatic life

17. What are the equipments used for chain surveying? (Nov 2018)

- Chain
- Tape
- Pegs
- Arrow or marking pins
- Ranging rods
- Cross staff
- Plumb bob

DETAIL QUESTIONS:

1. Classify the types of survey based on instruments used.(Jan 2015, Nov 2016, Nov 2017)
2. Explain the principle of leveling.(Nov 2011)

3. State the principles and uses of surveying.(May 2012)
4. Explain the construction and working principle of prismatic compass used in surveying.(April/May 2014, may 2019)
5. Define contours. Explain the characteristics and uses of contours.(May 2013, April/May 2014)
6. What is WBM road? Explain the construction of WBM road.(May 2013,Jan 2013)
7. (a) What are the component parts of water bound macadam road.(April 2011)

   (b) Discuss the advantages and disadvantages of water bound macadam road?(April 2010)
8. Discuss in detail the various classification of roads.(Nov2016)
9. State the advantages and disadvantages of concrete road.(Jan 2010)
10. With neat sketch explain the components of bridge. (April/May 2014, Jan 2010)
11. What is a bridge? What are meant by superstructure and substructure of a bridge? (Jan 2016, May 2016, Nov 2017)
12. What are the factors based which the site for dam is selected? Explain briefly.(Jan 2015,Jan 2016,nov 2016, April/May 2014)
13. What are the different methods of chlorination.(April 2010)
14. Write a critical note on the various types of roads. List their relative merits and limitations. (Nov 2018)
15. (a) Write a note on the types of bridges.

    (b) List the need for rain water harvesting. (Nov 2018)
16. Detail in construction features of gravity dam and earth dam with suitable sketch. (May 2019)